

Fraud Detection Without Specificity: A Null-Calibration Evaluation of the `eforensics` Model on Two Decades of Korean Elections

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Abstract

Bayesian finite-mixture election-forensics models in the tradition of Mebane [2016] classify vote-counting units as clean or fraudulent from aggregate turnout and vote-share data alone, without independent evidence of manipulation. We evaluate the `eforensics` quasi-binomial-logistic (`qbl`) model against 20 years of Korean elections (2002–2025), fitting 228 separate model runs spanning general and presidential elections, two aggregation levels, up to five vote-counting channels, and, critically, two independently constructed *fraud-free synthetic controls* calibrated to match each real dataset’s turnout and vote-share marginals without any fraud-generating mechanism. We find that the model reports substantial “fraud” (5–31% of units, up to 8.7 million “fraudulent” votes) on data that is fraud-free by construction; that real-data fraud-share estimates correlate at $r = 0.98$ with their fraud-free null counterparts across 58 paired cells; that the model’s largest real-data estimates fall on presidential elections held before Korea had early voting or any fraud allegations (2002–2012), exceeding what it reports for the actually contested 2020 election; and that recoding the model to test fraud in the opposite (conservative-favoring) direction reproduces the identical pathology, including assigning *more* false-positive fraud to the one pre-2014 election with a genuine historical manipulation allegation than to the allegation’s own claimed direction. We conclude that the model’s output depends on the shape of the (turnout, vote-share) distribution, not on fraud, and that this failure is not fixable by post-hoc recalibration, since recalibrating for validity would remove essentially all detected signal in the corpus.

Keywords: election forensics; Bayesian finite mixture models; null calibration; `eforensics`; early voting; South Korea; election fraud claims

1 Introduction

Bayesian election-fraud forensics models classify aggregation units (precincts, wards, administrative neighborhoods) into latent classes representing clean voting and one or more fraud mechanisms, using only reported vote totals and turnout. The most widely applied model in this tradition is Walter Mebane’s `eforensics` framework [Mebane, 2016, Ferrari et al., 2019], a finite mixture of overdispersed binomial models in which a subset of units are permitted to exhibit jointly elevated turnout and leader vote share, interpreted as evidence of vote manufacturing (converting abstentions into votes) and vote stealing (transferring votes from an opponent).

Korea offers an unusually clean natural experiment for auditing such a model. Nationwide in-person early voting (*sajeon tupyoo*, “pre-vote”) was introduced by statute in 2013 and first used in the 2014 local

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elections; the first general election held under the system was the 20th (2016) and the first presidential election was the 19th (2017). Before 2014, no election-fraud claim of the count-manipulation type this literature targets was raised about a Korean national election.¹ Organized fraud allegations first emerged after the 21st general election (April 2020), concentrated almost entirely on early-voting ballots, and Mebane himself applied `eforensics` to that election, reporting anomalies he characterized as strongly suggestive of manipulation, with roughly 28.7% of pre-vote (early-voting) units flagged as fraudulent [Mebane, 2020]. That analysis has since become a central citation for the Korean fraud-claim movement, which resurfaced around the 2024 general election and was cited in connection with the December 2024 martial-law declaration, followed by a snap presidential election in June 2025. A peer-reviewed Korean-language study, Min and Yoo [2021], independently shows that Mebane’s original data construction was invalid from the outset rather than merely a defensible alternative specification (Section 6).

This creates a sharp, falsifiable question. If `eforensics` is a genuine fraud detector, it should report approximately zero fraud on synthetic data constructed to share Korean elections’ turnout and vote-share distributions but containing no fraud-generating mechanism, and it should report approximately zero fraud on the pre-2014 Korean elections for which no fraud narrative exists. We test both predictions directly, run the identical analysis in both the historically alleged direction and its opposite, and find that the model fails both tests in a way that is fully explained by the shape of the (turnout, vote-share) distribution rather than by any fraud process.

2 The `eforensics` quasi-binomial-logistic model

The `eforensics` package’s `qbl` variant assigns each unit i , with eligible voters N_i , leader votes w_i , and abstentions a_i , to one of three latent classes $Z_i \in \{1, 2, 3\}$ (no fraud, incremental fraud, extreme fraud) with population mixture weights $\pi = (\pi_1, \pi_2, \pi_3)$. Conditional on class, w_i and a_i follow overdispersed binomial likelihoods with per-unit random effects on turnout (τ_i), leader share (ν_i), and, for fraudulent classes, fraud-magnitude parameters governing manufactured votes (converted from abstentions, via ι) and stolen votes (transferred from the opponent, via χ). Population-level location equations $w \sim \text{prov}$ and $a \sim \text{prov}$ allow province-level fixed effects. Estimation is by MCMC via JAGS. Throughout this study, every model fit calls the official `eforensics` R package [Ferrari et al., 2019] directly, installed from Walter Mebane’s own repository at the University of Michigan (`UMeforensics/eforensics_public` on GitHub, https://github.com/UMeforensics/eforensics_public), via its own `eforensics()` function; no part of the fraud model, the likelihood, the priors, or the MCMC sampler was reimplemented for this study. Our own code is limited to data preparation (constructing the unit-level tables in Section 3), the two null-data generators (Section 3.2), and post-processing of the package’s posterior output; the inferential machinery itself is exactly what a user obtains by installing and running the published package unmodified.

Two properties of this specification are essential to what follows. First, fraud is identified purely from the *shape* of the (turnout, vote-share) distribution: fraudulent units are those whose joint elevation of τ_i and ν_i is inconsistent with the fitted clean-class distribution. No information external to aggregate vote counts enters the model. Second, and less widely emphasized, fraud is **unidirectional**: both manufacturing and stealing mechanisms only ever add votes to whichever party or candidate is designated the “leader.” The model has no mixture component for fraud against the leader; votes lost by the leader are absorbed into the clean class regardless of their actual cause. Every published application of `eforensics` to a contested election therefore tests fraud in one direction only, chosen by the analyst.

¹One partial qualification concerns the 2012 presidential election: the 2017 documentary *The Plan* (Korean: *Deo Peullaen*) alleged that electronic ballot-sorting-machine misclassification (the “K=1.5” pattern) manipulated results in favor of the conservative winner, Park Geun-hye. This claim was statistically disputed at the time and never became a mass movement comparable to the post-2020 controversy; we return to it in Section 4.7.1, where it becomes directly useful.

3 Data and experimental design

3.1 Elections and aggregation grid

We assembled official National Election Commission results for eleven Korean national elections: five general (legislative) elections (18th [2008], 19th [2012], 20th [2016], 21st [2020], 22nd [2024]) and six presidential elections (16th [2002], 17th [2007], 18th [2012], 19th [2017], 20th [2022], 21st [2025]). The first five elections listed (18th/19th general, 16th/17th/18th presidential) predate early voting; all others were held under the early-voting system.

Each election was aggregated at two levels: *dong* (Korean administrative neighborhood, *beopjeong eupmyeondong*; 2,700–3,600 units after filtering) and *constituency* (*seongeogu*; 198–253 units). For early-voting-era elections, votes were further split by channel: at *dong* level, in-precinct early votes (*gwannae sajeon tupyoo*, “inside-precinct pre-vote”), same-day votes, and their pool; at constituency level, additionally out-of-precinct early votes (*gwano sajeon tupyoo*, “outside-precinct pre-vote,” observable only at this level) and combined totals. Pre-early-voting elections use a single “total, no early” channel. Units with fewer than 50 votes in any analyzed channel, or with zero leader votes across both early and same-day channels, were dropped; eligible-voter counts were reconstructed from registered-voter rows and floored at total votes cast.

The Democratic-lineage party or candidate was coded as the model’s “leader” in the primary analysis (Section 4), regardless of whether that party won; Section 4.7 reports a parallel analysis with the conservative-lineage side coded as leader. Model fits used JAGS with 4 chains, 1,000 burn-in and adaptation iterations, and either 3,000 or 5,000 iterations (two settings batches present in the run history for unrelated infrastructure reasons; convergence failure rates were statistically indistinguishable between them, 91% vs. 89%, so the split does not drive any result reported below).

3.2 Two fraud-free synthetic controls

The central methodological device of this study is a pair of synthetic datasets constructed to match each real dataset’s turnout and vote-share distributions while containing no fraud-generating process whatsoever.

Marginal null. For each unit i with real turnout τ_i and leader share ν_i , independently draw $\tau_i^* \sim \text{Beta}(k\tau_i, k(1 - \tau_i))$ and $\nu_i^* \sim \text{Beta}(k\nu_i, k(1 - \nu_i))$ with concentration $k = 200$, then reconstruct vote counts from τ_i^*, ν_i^* , and the unit’s true eligible-voter count N_i . This preserves every unit’s marginal turnout and share location, and hence all cross-unit heterogeneity in these two moments, while replacing whatever process actually generated the data with independent, clean Beta noise.

Joint null. Compute province-level means of ($\logit \tau$, $\logit \nu$) and the pooled 2×2 residual covariance across all units; simulate every unit as its province mean plus homogeneous bivariate Gaussian noise. This retains only province-level location and one global covariance structure; any unit-specific anomaly or fraud mechanism is destroyed by construction.

Both nulls are fit through the identical `eforensics` workflow, with identical filters and MCMC settings, as their paired real-data cell. The logic is straightforward: if `eforensics` reports substantial fraud on data that is fraud-free by construction, that is a measured false-positive rate; if it reports fraud on real data indistinguishable from what it reports on the paired null, its real-data output carries no information beyond the shape of the (turnout, share) marginal distribution that the null also reproduces.

Of 174 possible (election, level, channel, source) dem-coded cells, all 174 were fit to completion. A second grid with the conservative-lineage side coded as leader covers 54 cells: complete coverage of the 2012 presidential election, the 2020 general election, and the 2024 general election (Section 4.7).

4 Results

4.1 Massive false-positive fraud on data with no fraud

Across 116 completed null runs (marginal and joint nulls pooled), 40 (34.5%) report a fraud share (the posterior median of $\pi_2 + \pi_3$) above 5%, and 15 report above 15%. The largest false positives reach 31.4% of units (constituency-level total votes, 2025 presidential election, marginal null) with an estimated 8.66 million fraudulent votes. Table 1 lists selected examples.

Table 1: Selected fraud-free null runs with large false-positive fraud shares.

Cell (fraud-free null)	Fraud share	Fraudulent votes (F_w)	Units flagged
2025 pres., constituency, total, marginal	31.4%	8,663,935	34/242
2022 pres., constituency, total, marginal	27.1%	5,915,814	25/248
2017 pres., constituency, total, joint	23.4%	4,224,649	10/250
2012 pres., constituency, total, joint	20.3%	4,171,829	24/251
2012 pres., dong, total, marginal	17.5%	3,739,725	361/3479
2020 gen., constituency, total, marginal	14.1%	3,189,324	8/253

For calibration: Mebane’s 2020 analysis headlined 28.7% of early-voting units as fraudulent [Mebane, 2020]. Fraud-free synthetic data of the same country’s elections, matched only on turnout and share marginals, routinely produces flagged-unit rates and fraud shares of the same order of magnitude.

4.2 Real data is statistically indistinguishable from fraud-free data

Pairing each of the 58 real-data cells with its two null counterparts, real-data fraud share correlates at $r = 0.984$ with the marginal null and $r = 0.946$ with the joint null (Figure 1). Restricted to the early-voting channels at the center of the Korean controversy, the mean paired difference (real minus null) is under 0.2 percentage points. Real data exceeds *both* of its nulls in only 20 of 58 cells, roughly what sign noise alone would produce, and in every such case the excess is small relative to cross-null disagreement and occurs under severe non-convergence (Section 4.5).

4.3 The pre-early-voting natural experiment

Table 2 reports real-data results for the five elections held before early voting existed. Two patterns are decisive. First, the presidential elections of 2002, 2007, and 2012, for which no early-voting-type fraud allegation has ever been raised in Korea, receive fraud shares of 7–17%, with the 2012 presidential election (16.7–17.3%) exceeding every early-voting channel of the actually contested April 2020 election (3–7%; Section 4.6). If `eforensics`’ output were taken as evidence in the way it has been used for 2020, it would imply that 2002–2012 Korea was pervasively fraudulent and that 2020 was comparatively clean, the reverse of the claim structure the model has been used to support. Second, the general elections of the same era (2008, 2012) score near zero (0.5–0.7%), a roughly 25-fold swing driven by race structure (a single national presidential race with regionally polarized vote shares vs. many district-level party races) rather than by any fraud mechanism, exactly as the geometric account in Section 4.4 predicts. Both patterns are reproduced almost exactly by the paired fraud-free nulls of the same elections (e.g., the 2012 presidential nulls score 17.3–20.3%, matching the real estimate).

Figure 2 places these estimates on a timeline alongside the introduction of early voting and the onset of fraud claims: the model’s largest outputs fall precisely where no claim exists.

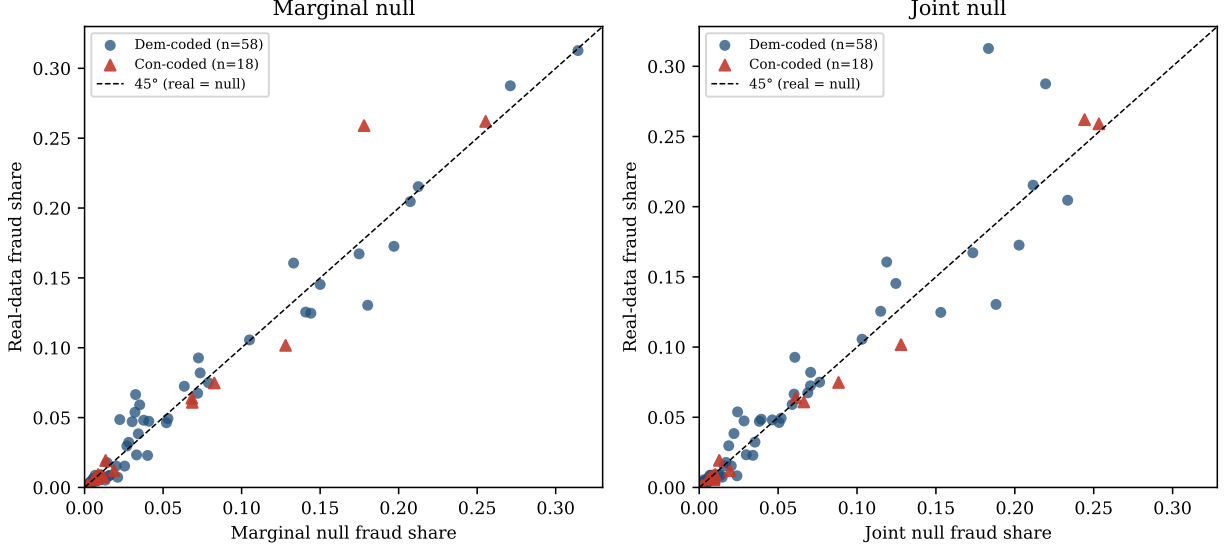


Figure 1: Real-data fraud share vs. fraud-free null fraud share, across 58 dem-coded cells (blue circles) and 18 conservative-coded cells with all three sources completed (red triangles). Points falling near the 45° line indicate that the model’s real-data output is statistically indistinguishable from what it reports on data constructed to contain no fraud.

4.4 The model signal tracks distributional geometry, not fraud

Mean real-data fraud share varies sharply and systematically by channel: same-day-only channels average 0.2–0.9%; early-only channels average 2–4%; pooled and total channels average 9–18%. Every one of these patterns is reproduced by the fraud-free nulls of the same channel. The mechanism is straightforward. Incremental fraud (π_2 , which dominates every large estimate in this corpus; extreme fraud π_3 is $\leq 0.3\%$ almost everywhere) is identified by units sitting in the high-turnout, high-share corner of the clean-class distribution. Per-channel turnout $\tau_i = \text{channel votes}_i / N_i$ is mechanically deflated in same-day-only channels (roughly 45–55%, since a substantial share of voters vote early) and even more deflated in early-only channels (25–35%), pulling both away from the high- τ corner. Pooling restores true turnout (65–80%) while simultaneously mixing two subpopulations with different (τ, ν) profiles, inflating overdispersion; the mixture model absorbs this overdispersion as “incremental fraud.” The pre-2014 presidential elections combine genuinely high turnout (70–76%) with extreme regional polarization of vote share (ranging from roughly 10% to over 90% across provinces), placing their single channel in the same danger zone. Because both null generators preserve exactly this turnout-and-share geometry and destroy everything else, they reproduce the real estimates almost perfectly. The quantity `eforensics` recovers in this setting empirically depends, smoothly and predictably, on where the (τ, ν) cloud sits and how fat its tails are, computable with certainty from data containing no fraud.

4.5 Convergence and the model’s own multimodality diagnostic

Of 174 dem-coded runs, 156 (89.7%) fail the standard Gelman–Rubin criterion ($\hat{R} \leq 1.05$) on the mixture weights; the median maximum $\hat{R}$ is 1.64, and 17 runs exceed $\hat{R} = 5$. Mebane [2023] proposes a specific diagnostic for posterior multimodality in this model class, $M(\pi_j)$, the largest cross-chain difference in chain-specific posterior means, combined with a Hartigan dip-test p -value on pooled draws, arguing that its presence signals substantive model misspecification (“lost votes”) rather than a mere numerical artifact.

Table 2: Real-data results for elections held before early voting existed.

Election	Level	Fraud share	Fraudulent votes (F_w)	Units flagged
Pres. 2002	Constituency	12.5%	1,960,598	14/244
Pres. 2002	Dong	14.5%	2,188,462	355/3515
Pres. 2007	Constituency	8.2%	1,218,101	11/248
Pres. 2007	Dong	7.2%	882,177	165/3576
Pres. 2012	Constituency	17.3%	3,602,441	21/251
Pres. 2012	Dong	16.7%	3,236,794	291/3479
Gen. 2008	Constituency	0.7%	0	0/198
Gen. 2008	Dong	0.5%	17,914	2/2733
Gen. 2012	Constituency	0.7%	0	0/209
Gen. 2012	Dong	0.5%	22,770	4/2923

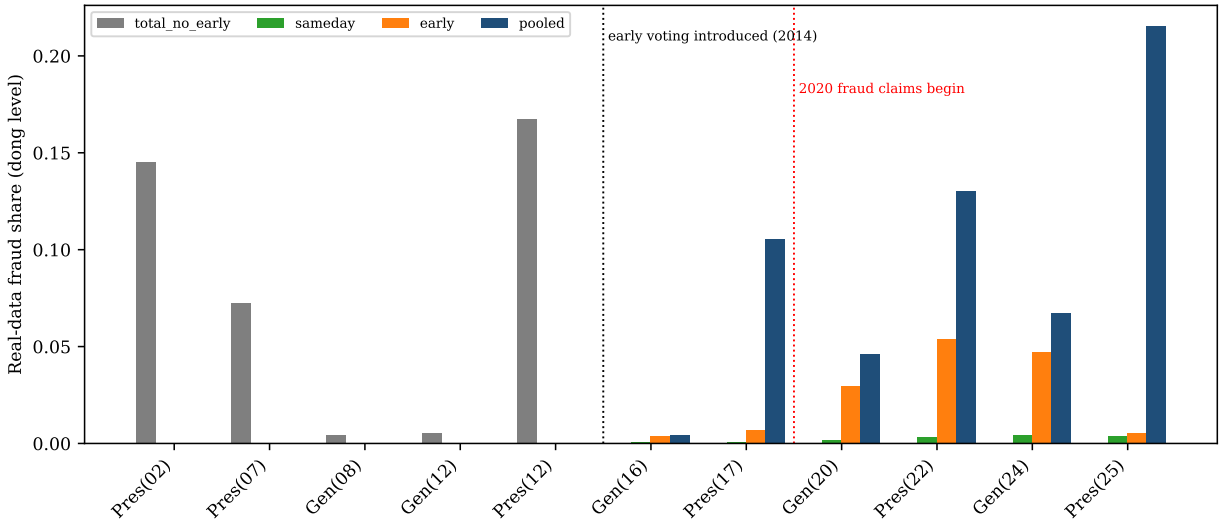


Figure 2: Real-data fraud share by election and channel (dong level), in chronological order. Dashed lines mark the introduction of early voting (2014) and the onset of organized fraud claims (2020). The largest estimates fall in the era with no claims; same-day-only channels remain near zero throughout.

This flag fires on 15/58 (25.9%) of real-data runs and 27/116 (23.3%) of fraud-free null runs: a diagnostic that fires at statistically indistinguishable rates on data with and without fraud carries no evidential weight about fraud, whichever interpretation of its cause is correct.

4.6 The contested election itself: April 2020

For the election that launched the Korean controversy and that Mebane [2020] analyzed directly: dong-level in-precinct early votes score a real fraud share of 2.97% (12/3480 units flagged, $\hat{R} = 2.65$, not converged), against fraud-free nulls of 2.69% and 1.87%. Constituency-level early votes score 6.65% against nulls of 3.25% and 6.01%, one of the few cells where real exceeds both nulls, addressed further in Section 4.7. Out-of-precinct early votes, the specific stream most heavily targeted by fraud claims, score 3.23%, falling *between* its two fraud-free controls (2.81%, 3.53%). Nothing in the 2020 early-vote channels stands out against the model’s own false-positive baseline, and every 2020 estimate is dwarfed by what the model incorrectly reports for the uncontested 2002–2012 elections or for fraud-free pooled data generally.

4.7 Bidirectional confirmation: fraud for whichever side is coded as leader

Because `eforensics` fraud mechanisms are unidirectional (Section 2), every result above tests only Democratic-favoring fraud. We reran the identical analysis, using the same data, the same filters, the same null generators, and the same MCMC settings, with the conservative-lineage candidate or party coded as leader instead, for 54 cells: the full 2012 presidential election (both levels), the full 2020 general election (both levels, every channel), and the full 2024 general election (both levels, every channel).

4.7.1 The 2012 presidential election, revisited

The 2012 presidential election is the one pre-2014 election with an actual historical count-manipulation allegation, and that allegation (*The Plan*, Section 1) claimed manipulation *favoring the conservative winner*. Table 3 and Figure 3 report both codings side by side.

Table 3: 2012 presidential election, dem- vs. conservative-coded leader.

Level	Leader coding	Source	Fraud share	$\hat{R}$
Dong	Dem	Real	16.7%	4.50
Dong	Dem	Marginal null	17.5%	2.04
Dong	Dem	Joint null	17.3%	1.44
Dong	Con	Real	25.9%	2.35
Dong	Con	Marginal null	17.8%	9.33
Dong	Con	Joint null	25.3%	9.82
Constituency	Dem	Real	17.3%	1.84
Constituency	Dem	Marginal null	19.7%	2.37
Constituency	Dem	Joint null	20.3%	1.62
Constituency	Con	Real	26.2%	1.24
Constituency	Con	Marginal null	25.5%	3.50
Constituency	Con	Joint null	24.4%	3.24

The model reports *more* fraud favoring the conservative winner ($\sim 26\%$) than fraud favoring the Democratic loser ($\sim 17\%$) in the identical election, and the conservative-coded estimate is, if anything, more tightly matched by its own nulls (constituency: real 26.2% bracketed by nulls of 24.4% and 25.5%; dong: real 25.9% nearly identical to the joint null’s 25.3%). Convergence is worse under conservative coding (maximum $\hat{R}$ up to 9.82, vs. 4.50 under Democratic coding). This is close to a best-case confirmation of the geometric account: pointed at either side of a regionally polarized, high-turnout, pre-early-voting election, `eforensics` reports large “fraud” favoring *that* side, matched almost exactly by fraud-free data with the same shape, and it does so more strongly in the direction with no evidentiary basis than in the direction with the (statistically disputed, but at least historically real) allegation.

4.7.2 The 2020 election, both directions

Conservative-coded estimates for the 2020 general election are smaller than Democratic-coded estimates in seven of eight compared channels (e.g., constituency total: 7.5% vs. 12.6%; dong early: 0.5% vs. 3.0%), directionally consistent with what the allegations claimed. This asymmetry, however, is exactly what the geometric mechanism predicts even under zero fraud: 2020 early voting, introduced amid the COVID-19 pandemic, skewed toward younger and more urban voters and hence toward the Democratic side on both real and null data alike. Consistent with a non-fraud explanation, the conservative-coded real values remain closely matched by their own conservative-coded nulls in every one of these channels.

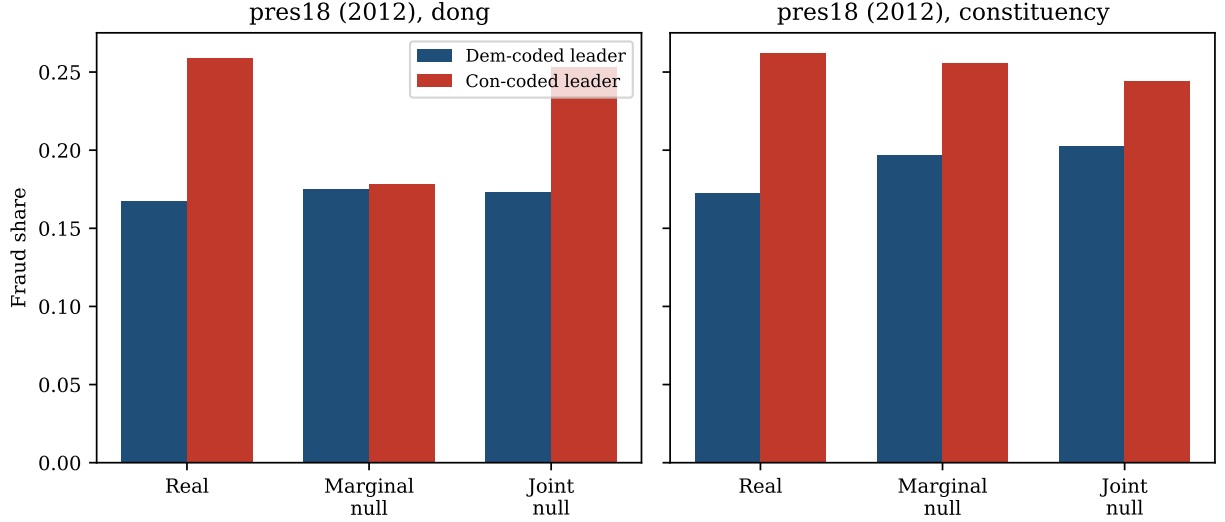


Figure 3: 2012 presidential election fraud share under Democratic-coded (blue) and conservative-coded (red) leader, real data and both fraud-free nulls, at dong and constituency level. The conservative-coded estimate is larger than the Democratic-coded estimate under every source, and both are closely tracked by their respective fraud-free nulls.

One cell is doubly informative. The constituency-level early-vote channel is one of the few cells (Section 4.2) where Democratic-coded real data (6.65%) exceeds both its nulls (3.25%, 6.01%); the conservative-coded version of the *same* channel shows the identical qualitative pattern (real 1.93% vs. nulls 1.34%, 1.27%). If this excess reflected genuine Democratic-favoring fraud, it should not simultaneously appear as excess conservative-favoring signal under the opposite coding, since the two runs use entirely different vote columns from the same underlying counts; true fraud for one side does not mechanically inflate the model’s estimate of fraud for the other. That the anomaly appears under both codings points instead to a structural feature of this specific channel’s real (τ, ν) distribution that neither null generator fully reproduces: a data-geometry gap, not a fraud signal.

4.7.3 The 2024 election, both directions

The 2024 general election, the most recent Korean election with an active fraud narrative, is now covered at both levels and every channel under conservative coding. The asymmetry seen in 2020 recurs and is, if anything, more consistent: conservative-coded real-data estimates are smaller than Democratic-coded estimates in all eight compared channels (Figure 4 shows the two largest, constituency total and dong pooled), for example constituency total (10.2% vs. 16.1%) and dong early (0.7% vs. 4.7%). Unlike 2020, none of the eight conservative-coded 2024 trios has real exceeding both of its nulls; in several channels, including constituency total (real 10.2% against nulls of 12.8% and 12.8%) and constituency early (real 1.2% against nulls of 1.9% and 1.9%), real data falls *below* both fraud-free controls. The one channel where the two codings converge most closely is dong pooled (conservative 6.4% against a Democratic-coded 6.7%, with conservative-coded nulls of 6.1–6.8% bracketing the conservative-coded real value), the same channel that showed the closest cross-coding agreement in 2020. As with 2020, the dem/con asymmetry is naturally read as reflecting the real, non-fraudulent demographic composition of early voters rather than a fraud signal, and the absence of any both-nulls-exceeded cell here removes even the single ambiguous case that 2020 provides.

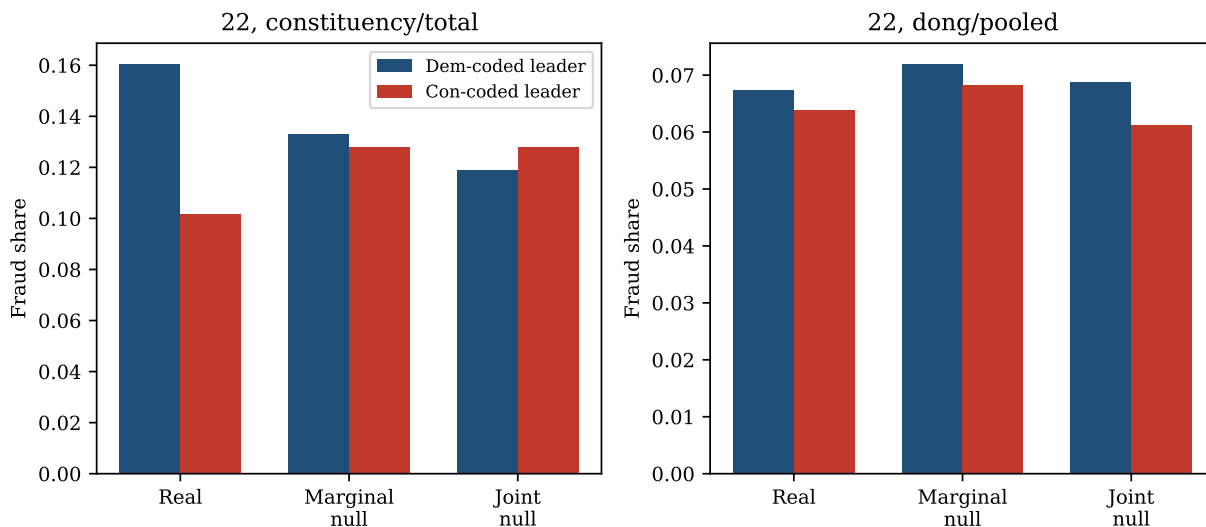


Figure 4: 2024 general election fraud share under Democratic-coded (blue) and conservative-coded (red) leader, real data and both fraud-free nulls, for the two largest-signal channels. The conservative-coded real estimate sits at or below its own nulls in both panels.

Convergence and the Mebane multimodality flag replicate across the full conservative-coded grid: 53/54 (98.1%) fail $\hat{R} \leq 1.05$, and the multimodality flag fires on 8/54 (14.8%) of runs, broadly consistent with the 15–26% rates found on both real and null data under Democratic coding.

5 Discussion: is eforensics credible as a fraud detector?

The evidence supports five independent, mutually reinforcing conclusions against treating eforensics’s positive outputs as evidence of fraud in this setting.

It has no specificity. It assigns 5–31% fraud shares and millions of fraudulent votes to data that is fraud-free by construction, under two distinct clean data-generating processes. A detector whose false-positive output matches its real-data output in magnitude detects nothing.

It fails its historical control. The Korean fraud narrative is inseparable from early voting (introduced 2014; claims from 2020). The model’s largest real-data estimates fall on pre-2014 presidential elections with no fraud narrative, exceeding its estimates for the actually contested 2020 channels. The temporal pattern of its output is uncorrelated with, and in places anti-correlated with, the temporal pattern of the allegations it has been used to support.

Its output is predictable from fraud-irrelevant geometry. Fraud share rises monotonically with channel-level turnout and cross-unit (τ, ν) heterogeneity, a pattern fully reproduced under both nulls (Section 4.4). The estimand the model recovers in practice is overdispersion-plus-corner-mass, not fraud.

Its inferential machinery is unreliable in exactly this application. 89.7% of runs fail conventional convergence diagnostics, and the model’s own advanced multimodality diagnostic fires at statistically indistinguishable rates on fraud-free and real data.

It is not directionally selective. Coding the opposite party as leader reproduces the identical pathology (Section 4.7), and in the one election with an actual historical manipulation allegation, assigns more false-positive fraud to the alleged direction’s true target than to the unalleged direction, the reverse of what a genuine detector calibrated to that allegation would produce.

5.1 Is the model salvageable by recalibration?

It is useful to locate this failure precisely rather than dismiss the model class wholesale. Failures of this kind fall on a three-level ladder: (i) *miscalibration*, fixable by recalibrating thresholds against a reference null distribution; (ii) *lack of power after calibration*, where size can be corrected but no signal remains; (iii) *non-identification of the estimand*, where the quantity recovered is not the one of interest regardless of calibration. Our evidence places `eforensics`-as-deployed at (iii) shading into (ii): fraud classes are identified purely through parametric shape assumptions on the clean class, so realistic clean-data heterogeneity (fat tails, high-turnout polarized units, multi-mode pooled voting) is absorbed into the fraud components, which is precisely why fraud-free nulls reproduce the real-data estimates. Recalibrating thresholds against an empirical null distribution (Section 7) would repair validity, since the false-positive rate could in principle be controlled, but because real and null outputs are already statistically indistinguishable, a validity-correct recalibrated test would report approximately no detectable fraud anywhere in this 20-year corpus: calibration fixes size, not power, and would empty every positive finding the model has previously been used to support. Genuine repair would require redesign: a flexible clean-class distribution plus injection-verified power demonstrating that realistic fraud magnitudes remain detectable above a corrected false-positive floor, with the live possibility that a sufficiently realistic clean class leaves nothing detectable, implying that aggregate-count-only mixture forensics of this type is underpowered in principle for high-turnout, regionally polarized electorates such as Korea’s, rather than merely misapplied in this instance.

We note explicitly that Mebane’s own work cautions that `eforensics` “frauds” may reflect strategic behavior rather than malfeasance [Mebane, 2020]. Our findings sharpen this caveat considerably: the estimates do not merely admit benign explanations, they are quantitatively reproduced by explicitly fraud-free processes matched only on distributional shape, in both the alleged and the unalleged direction. Taken seriously, this caveat implies the model’s outputs should not have been treated as evidence in the Korean controversy.

6 Limitations

Several limitations should be weighed against the conclusions above. Our nulls are single-seed draws (Section 3.2); the tightness of real-null agreement across 58 independent cells makes seed noise an implausible sole driver, but a formal ensemble is needed to attach calibrated p -values (Section 7). Our aggregation choices (dong/constituency level, province fixed effects) differ from Mebane [2020]’s precinct-level units and covariates, but this should not be read as a mere difference in reasonable specifications. Min and Yoo [2021] independently show that Mebane’s original units were not administratively valid for Korean election data: his “district, election-day” and “voting-post, election-day” units duplicate each other, and, more consequentially, his early-vote turnout was computed against a “registered voters” denominator that does not exist as a fixed population for Korean in-precinct early voting, which has no pre-registration, unlike the U.S. case his software assumes. This produces measured early-vote turnout clustering near 100% by construction, which mechanically inflates `eforensics`’ manufactured-vote channel specifically, since that channel is triggered by turnout appearing implausibly high. Correcting this error, Min and Yoo [2021] replicate the identical `eforensics-qbl` model and find Mebane’s headline fraud estimate falls by roughly half to two-thirds. The Democratic-lineage party is coded as leader throughout our primary analysis regardless of whether it won a given election; Section 4.7 addresses this directly for the 2012 presidential election and the 2020 and 2024 general elections in full, but the conservative-coded grid does not yet mirror the complete 174-cell dem grid, so bidirectional claims about elections not yet rerun under conservative coding (the 2002, 2007, and 2022 presidential elections, the 2017 presidential election, and the pre-2014 general elections) should be read as predictions consistent with the geometric mechanism rather than established results. Finally, 89.7% of runs fail conventional MCMC convergence; while non-convergence is symmetric between

real and null runs (making every real-vs-null comparison settings-matched) and a settings quasi-experiment embedded in our run history (68 runs at lower iteration count vs. 105 at higher) shows no systematic difference in outcomes, a small number of substantially longer targeted reruns would further close this objection (Section 7).

7 Further work

We highlight the additions most likely sharpen the conclusions above, in priority order. **(1) Null ensembles with calibrated p -values:** refitting each null generator with ≥ 50 seeds per cell would convert the paired-difference observation of Section 4.2 into a formal hypothesis test. **(2) A model-consistent self-null:** simulating from the fitted `qbl` model itself with π forced to $(1, 0, 0)$ would isolate whether false positives stem specifically from clean-class misspecification. **(3) An injection/recovery power study:** adding known fraud at realistic magnitudes (0.5–5% manufactured share in 1–10% of units) to null data and measuring recovery above the false-positive floor is necessary before any positive `eforensics` finding can be considered interpretable, in this application or elsewhere. **(4) Targeted long-chain reruns** of a small number of headline cells ($n.\text{iter} \geq 50,000$, 8 chains) would close the residual objection that non-convergence alone explains the findings, though existing evidence makes this unlikely. **(5) Completing the conservative-coded grid**, particularly the 2007 and 2022 presidential elections, would convert Section 4.7’s result into a fully complete bidirectional comparison. **(6) External placebo elections** from a country with no fraud discourse, run through the identical procedure, would provide a cross-national control.

8 Conclusion

Across 228 model fits spanning two decades of Korean elections, two independent aggregation levels, up to five vote-counting channels, two fraud-free synthetic controls, and two directions of leader coding, the `eforensics` quasi-binomial-logistic model’s fraud estimates are statistically indistinguishable from what it reports on data engineered to contain no fraud, are largest precisely where no fraud allegation has ever existed, and reproduce identically when the model is pointed at the opposite side of a contested election, including, in the one election with a genuine historical manipulation allegation, assigning more spurious fraud to the allegation’s true target than to its opposite. The model’s output is well explained, in every instance we examined, by the shape of the turnout-and-vote-share distribution alone. We do not conclude that Korean elections have been free of any irregularity; we conclude only that this model, as deployed, provides no evidence on the question either way, and that its positive findings in the public record should not be treated as evidence of fraud.

Data availability

The data, codes and results are available at github page and repository:

https://mkimacad.github.io/south_korean_election_forensics.

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